Name _____ Period _____ Date _____

Where Did the Old Ocean Floor Go?

Focus question: How old is the ocean floor compared with the continents, and how do moving plates make and destroy it?

Hook data

UNIT 3.1 · DAY 1 · GLUE IN · REFERENCE **Hook data: ocean floor and continent ages**

Place on the ocean floor	Age of the rock
Right at the Mid-Atlantic Ridge	0 (forming now)
Middle of the Atlantic, about 2,000 km from the ridge	about 80 million years
Atlantic seafloor next to the east coast of North America	about 180 million years
Right at the East Pacific Rise	0 (forming now)
Western Pacific, east of Japan (the oldest on Earth)	about 180 million years

Continental material	Where	Age
Oldest mineral (a zircon crystal)	Jack Hills, Western Australia	about 4.4 billion years
Oldest known rock (Acasta Gneiss)	Northwest Territories, Canada	about 4.0 billion years
Very old rock (Isua belt)	Greenland	about 3.8 billion years
Oldest widely accepted fossils (stromatolites)	Western Australia	about 3.5 billion years

Notice and wonder

I notice the ocean floor is much younger than the continents. I wonder where the old ocean floor went.

Vocabulary

Word	Definition (my words)	This reminds me of… (connections)
lithosphere (plate)	A rigid slab of Earth's outer layer that moves.	The Pacific plate
plate boundary	Where two plates meet.	San Andreas Fault
mid-ocean ridge	Where new ocean crust forms.	Mid-Atlantic Ridge
subduction	One plate sinks under another.	Nazca plate
trench	Deep groove where a plate sinks.	Mariana Trench

Name _____ Period _____ Date _____

Where Did the Old Ocean Floor Go? (continued)

Unit CER: glue U-1 on its own new page and write your first claim.

Plot the pattern

Glue in: your plotted world map. Keep the site list (D1-A) beside it.

Sample: points in lines along the Pacific rim and the mid-Atlantic

Pattern sentence

The points make lines. They match the edges of the plates.

Plate boundary models

Sketch divergent, convergent, and transform. Label the arrows.

Sample: arrows apart with a ridge; a blue slab under a tan block with a trench; arrows sliding past each other

Boundary table

UNIT 3.1 · DAY 1 · GLUE IN · WRITE ON IT **Plate boundary table**

Fill in the table after you make each clay model. Divergent is done for you.

Boundary	What the plates did	What formed	Real example
Divergent	Moved apart	New crust at a mid-ocean ridge	Mid-Atlantic Ridge
Convergent	Pushed together; the ocean slab slid under	A trench and volcanoes (subduction)	Pacific coast of South America
Transform	Slid past each other	A broken line; earthquakes	San Andreas Fault

Exit ticket

Ocean crust is made at mid-ocean ridges and sinks back down at trenches (subduction).

Name _____ Period _____ Date _____

Reading Rock Ages

Focus question: What patterns do the ages of ocean crust and continental crust follow?

Warm-up

The Brazil sample is older. It is far from the ridge, and crust gets older away from the ridge.

Seafloor ages

UNIT 3.1 · DAY 2 · GLUE IN · REFERENCE **Seafloor ages and color key**

Distance from ridge (km)	Crust age (million years)
0	0
500	20
1,000	40
1,500	60
2,000	80
2,500	100
3,000	120

Age (million years)	Color
0	red
20	orange
40	yellow
60	green
80	blue
100	purple
120	dark purple

UNIT 3.1 · DAY 2 · GLUE IN · WRITE ON IT **Seafloor strip**

Color each box using the key on D2-A. The ridge is done for you.

3,000 km	2,500	2,000	1,500	1,000	500	RIDGE	500	1,000	1,500	2,000	2,500	3,000 km
dark purple	purple	blue	green	yellow	orange	red	orange	yellow	green	blue	purple	dark purple

As you move away from the ridge, the crust gets **older** .

Vocabulary

Word	Definition (my words)	This reminds me of… (connections)
craton	A continent's ancient core.	Canadian Shield

Name _____ Period _____ Date _____

Reading Rock Ages (continued)

Graph: age vs. distance

UNIT 3.1 · DAY 2 · GLUE IN · WRITE ON IT **Graph grid: age vs. distance**

Mark a dot in the box where the distance column meets the age row. Connect the dots. The first dot (0 km, 0 years) is done for you.

Age (million years)	0 km	500	1,000	1,500	2,000	2,500	3,000 km
120							●
100						●	
80					●		
60				●			
40			●				
20		●					
0	●						

Rate = distance ÷ time = 25 km per million years = 2.5 cm per year. Age at 3,500 km: 140 million years.

Color North America

UNIT 3.1 · DAY 2 · GLUE IN · REFERENCE AND WRITE ON IT **North America crust ages and strip**

Color the oldest region dark brown and the youngest light yellow. Choose three colors in between. The Canadian Shield is done for you.

Region of North America	Direction and distance from the central core (km)	Age of the crust
Pacific Coast Ranges	west, about 2,800	about 50 million years
Rocky Mountain region	west, about 1,800	about 150 million years
Canadian Shield (the central core)	0	about 2.7 billion years
Grenville Province (Great Lakes to Quebec)	east, about 1,200	about 1.1 billion years
Appalachian Mountains	east, about 2,000	about 400 million years

Pacific Coast Ranges (west)	Rocky Mountains	Canadian Shield (core)	Grenville Province	Appalachians (east)
light yellow	yellow	dark brown	brown	tan

Add two rows to your Unit CER page (U-1): the ocean pattern and the continent pattern.

Name _____ Period _____ Date _____

Reading Deep Time and Early Earth

Focus question: How do scientists know how old Earth is, and why do meteorites and moon rocks tell Earth's age better than Earth rocks do?

Warm-up

Use atoms that change at a steady rate, like a clock, to date rocks.

Vocabulary

Word	Definition (my words)	This reminds me of… (connections)
impact crater	A dent made by a rock hitting a surface.	Moon highlands

Crater counts

UNIT 3.1 · DAY 3 · GLUE IN · WRITE ON IT **Crater counts**

Count the craters at each time step in the simulation. Row 1 is done for you. Then answer the questions.

Time step	Moon craters	Earth craters
1 billion years	6	5
2 billion years	12	4
3 billion years	18	6
4 billion years	24	5

Which surface looks older? How can you tell?

The Moon: it has many craters that stay.

Why does Earth show so few craters?

Erosion, volcanoes, and moving plates erase them.

A surface has a lot of craters. Is it old or young? Why?

Old: craters pile up over time.

Name _____ Period _____ Date _____

Reading Deep Time and Early Earth (continued)

45 cm Earth timeline

UNIT 3.1 · DAY 3 · GLUE IN · REFERENCE AND WRITE ON IT **Timeline table and old materials**

Scale: 1 cm = 100 million years. Position = (4,500 − age in millions of years) ÷ 100. Two rows are done for you.

Event	Age (millions of years ago)	Position on the 45 cm strip (cm)
Earth forms (left end)	4,500	0
Oldest mineral on Earth (zircon)	4,400	1
Oldest known rock	4,000	5
Oldest fossils	3,500	10
Oldest ocean floor	200	43
Today (right end)	0	45

Material	Where	Age
Meteorite fragments (Canyon Diablo)	Arizona, USA	about 4.56 billion years
Moon rocks from the lunar highlands	Brought back by Apollo	up to about 4.4 billion years
Oldest Earth mineral (zircon)	Jack Hills, Australia	about 4.4 billion years
Oldest known Earth rock (Acasta Gneiss)	Canada	about 4.0 billion years
Oldest widely accepted fossils (stromatolites)	Western Australia	about 3.5 billion years
Oldest ocean floor	Western Pacific and Atlantic margins	about 180–200 million years

Your timeline strip

Tape your 45 cm strip here and label each event.

Sample: strip marked at 0, 1, 5, 10, 43, and 45 cm

Exit ticket

Meteorites and moon rocks have barely changed, so they keep the early record; Earth's early rocks were recycled.

Add a row to your Unit CER page (U-1) about meteorites, moon rocks, or zircons.

Name _____ Period _____ Date _____

Why the Ocean Is Young and the Continents Old

Focus question: How is ocean crust made, grown, and recycled, and how do continents grow and survive?

Warm-up

It sinks back into the mantle at a trench (subduction).

Guided notes

UNIT 3.1 · DAY 4 · GLUE IN · WRITE ON IT **Guided notes**

Fill in the blanks as your teacher goes. Blank 1 is done for you.

1. Old ocean crust is made at a **mid-ocean ridge** and moves away about 2.5 cm a year (25 km per million years).
2. A piece of crust travels 3,000 km. Time = 3,000 ÷ 25 = 120 million years. In 200 million years it travels 200 × 25 = 5,000 km, so it reaches a trench and sinks. No ocean floor is older than about 200 million years.
3. Ocean crust has a density of about **3.0 g/cm³**. Continental crust is about **2.7 g/cm³**. The ocean plate sinks (*subduction*).
4. Each continent has an ancient core. Volcanic arcs and pieces of crust are added to its edge over time, like rings on a tree. The edge is younger than the core.

Sink or float

UNIT 3.1 · DAY 4 · GLUE IN · WRITE ON IT **Density table**

Water has a density of 1.0 g/cm³. Density = mass ÷ volume.

Block	Mass (g)	Volume (cm³)	Density (g/cm³) = mass ÷ volume	Sinks or floats?
Wood block (stand-in for a continent)	40	64	0.6	floats
Clay block (stand-in for an old, cold ocean plate)	128	64	2.0	sinks

Which block is denser? clay What does the denser one do when they meet? dives under

Sample values; real numbers depend on your blocks.

Name _____ Period _____ Date _____

Why the Ocean Is Young and the Continents Old (continued)

SketchNotes checklist

UNIT 3.1 · DAY 4 · GLUE IN · WRITE ON IT **SketchNotes checklist**

Check each box when it is in your SketchNotes. Turn in a photo on Canvas: Lesson 4: SketchNotes.

	Half page of notes		Cross-section drawing
☑	Where ocean crust is made	☑	Ridge and plate moving away
☑	How it moves and sinks	☑	Trench and volcanic arc
☑	How continents grow from a core	☑	Continent with an ancient core and a younger edge
☑	Density numbers	☑	Density labels and a carbon arrow

SketchNotes: notes

Half a page: the life of ocean crust and how continents grow.

Sample: crust is made at a ridge, moves about 2.5 cm a year, and sinks at a trench in under 200 million years; continents grow outward from an ancient core

Sensemaking: cross-section

Draw a ridge, moving plate, trench, volcanic arc, and a continent with a younger edge. Label the densities and add a carbon arrow.

Sample: ridge, plate arrows, trench, volcanic arc, continent with a younger edge, 3.0 and 2.7 g/cm^3 labels, carbon arrow down and back up

Add a row to your Unit CER page (U-1) about density or continental growth.

Name _____ Period _____ Date _____

Evidence for Moving Crust

Focus question: How strong is the evidence that the crust has moved and is still moving?

Warm-up

Past: rock ages, stripes, and fossils. Today: GPS and the Hawaiian islands.

Vocabulary

Word	Definition (my words)	This reminds me of… (connections)
magnetic reversal	The magnetic field flips; crust records it.	Mirror stripes

Magnetic stripes

UNIT 3.1 · DAY 5 · GLUE IN · REFERENCE **Magnetic stripe bands**

Color both strips with this table (2 cm = 1 million years). Start at the end you will push through first.

Band (from the end you push first)	Color	Length	Age it represents
1	Reversed (blue)	3.4 cm	about 3.6–5.3 million years
2	Normal (red)	2.0 cm	about 2.6–3.6 million years
3	Reversed (blue)	3.6 cm	about 0.8–2.6 million years
4	Normal (red)	1.6 cm	0–0.8 million years (youngest)

Tape in your paper stripes

Tape the stripes here. Label the youngest and oldest crust.

Sample: mirror-image red and blue stripes on both sides of the slit; youngest at the center

Name _____ Period _____ Date _____

Evidence for Moving Crust (continued)

Hawaii and GPS

UNIT 3.1 · DAY 5 · GLUE IN · REFERENCE AND WRITE ON IT **Hawaii and GPS**

1 km per million years = 0.1 cm per year. GPS today: about 7 to 8 cm per year. The first rate is worked for you.

Volcano or island	Distance northwest of Kīlauea (km)	Age of its oldest rock (million years)
Kīlauea (Hawai'i Island, active now)	0	0
Kaua'i	about 500	about 5
Midway Atoll	about 2,500	about 28

Kīlauea to Kaua'i: 500 ÷ 5 = **100** km per million years = **10** cm per year (done for you).

Kīlauea to Midway: 2,500 ÷ 28 = about 90 km per million years = about 9 cm per year.

Pangaea puzzle

Tape your best fit here.

Sample: South America fits Africa; Mesosaurus and Glossopteris marked on both

UNIT 3.1 · DAY 5 · GLUE IN · REFERENCE **Fossil table**

Organism	What it was	Found in
Mesosaurus	freshwater reptile	South America and southern Africa
Glossopteris	seed fern (heavy seeds)	South America, Africa, India, Australia, Antarctica
Lystrosaurus	land animal	Africa, India, Antarctica

Evaluate the evidence

UNIT 3.1 · DAY 5 · GLUE IN · WRITE ON IT **Evaluate the evidence**

Rate strength from 1 (weak) to 4 (very strong). Row 1 is started for you.

Line of evidence	Measured numbers?	Past or present motion?	Independent of the others?	Strength (1–4)
Rock ages across a ridge	Yes	Past	Yes	4
Magnetic stripes	Yes	Past	Yes	4
GPS and Hawaiian island ages	Yes	Present and past	Yes	4
Matching fossils and rocks	No	Past	Partly	2 to 3

Name _____ Period _____ Date _____

Evaluate the Theory, Tell Earth's Story

Focus question: How well does plate tectonics explain the ages of crustal rocks, and what story do ancient materials tell about Earth's formation?

Sort your evidence

UNIT 3.1 · DAY 6 · GLUE IN · WRITE ON IT **Three buckets**

Put each row of your Unit CER log in a bucket. One example is done for you.

The seafloor is young because…	The continents are old because…	The early record is missing because…
The oldest ocean floor is only about 200 million years old (done for you).		
Crust is 0 at the ridge and 120 million years at 3,000 km.	The Canadian Shield is about 2.7 billion years old.	Plate tectonics and erosion recycled it.
Ocean plates sink at trenches.	Continents are less dense, so they stay on top.	Scientists use meteorites and moon rocks.

Part A: argument

UNIT 3.1 · DAY 6 · GLUE IN · REFERENCE **Part A sentence frames**

1. **Claim:** Plate tectonics explains the ages of Earth's crustal rocks (very well / partly / poorly) because _____.
2. **Ocean evidence:** The seafloor is youngest at the _____ and gets older with _____; the oldest ocean floor is only about _____ million years old.
3. **Continent evidence:** Continental crust as old as _____ exists, and in North America the crust gets _____ toward the edges.
4. **Reasoning:** Because ocean plates are _____ than continental crust, they _____ at subduction zones, while continents _____.
5. **Limit or counterclaim:** One thing this evidence cannot show is _____. OR: Someone might say the ocean floor is young because _____. This does not fit the evidence because _____.

Write your argument

Plate tectonics explains the ages very well. The seafloor is youngest at the ridge (0) and no older than about 200 million years, while continents have crust up to 4.4 billion years old. Ocean plates are denser, so they sink at trenches, while continents survive. One limit: no early ocean floor is left, so we also use meteorites.

Name _____ Period _____ Date _____

Evaluate the Theory, Tell Earth's Story (continued)

Peer feedback

UNIT 3.1 · DAY 6 · GLUE IN · WRITE ON IT **Peer feedback**

Trade papers. Give one star (something strong) and one step (one thing to improve).

Partner's name	One star	One step
Sam	You used the number 2.5 cm per year.	Say what the evidence cannot show.

Part B: Earth's story

UNIT 3.1 · DAY 6 · GLUE IN · WRITE ON IT **Evidence cards**

Cut the cards apart. Put them in order from oldest to newest on a paper strip and glue them down. Card 1 is placed for you.

Order	Evidence card	What it shows
1	Meteorites: about 4.56 billion years	The solar system formed about 4.56 billion years ago.
2	Moon highland rocks: up to about 4.4 billion years; heavily cratered	Solid crust existed early; the early solar system had many impacts.
3	Earth's oldest zircons: about 4.4 billion years	Solid crust and minerals had formed on Earth by then.
4	Oldest Earth rock (Acasta Gneiss): about 4.0 billion years	Rock has survived for 4 billion years in the continents.
5	Oldest fossils (stromatolites): about 3.5 billion years	Life existed by then.
6	Oldest ocean floor: about 200 million years	Ocean crust is recycled; the early record is not on the ocean floor.

Story frames

UNIT 3.1 · DAY 6 · GLUE IN · WRITE ON IT **Part B story frames**

Use your ordered evidence cards. Frame 1 is done for you.

1. About **4.56** billion years ago, **the solar system formed**. The evidence is **meteorites**.
2. About 4.4 billion years ago, solid crust and zircon minerals formed. The evidence is moon rocks and zircons.
3. About 3.5 billion years ago, life existed. The evidence is stromatolite fossils.
4. Earth's earliest record is missing because plate tectonics and erosion recycled it.

Mark every row of your Unit CER page (U-1) that you used today.

Name _____ Period _____ Date _____

Review Day

Focus question: Which targets do I still need to work on before the CSA?

Self-rating chart

UNIT 3.1 · DAY 7 · GLUE IN · WRITE ON IT **Self-rating chart**

Rate each target from 1 (I need help) to 4 (I could teach it). Use your CFA scores to check.

Target	My rating (1–4)	My CFA score	What I will do today
ESS1-5.1: I can describe Earth's outer layer as moving plates, name three kinds of plate boundaries, and describe where ocean crust is made and where it sinks back down.	(personal)	(personal)	(personal)
ESS1-5.2: I can read a rock-age map and describe the pattern in the ages.	(personal)	(personal)	(personal)
ESS1-5.3: I can explain why ocean crust is young and why continents are oldest at their ancient cores.	(personal)	(personal)	(personal)
ESS1-5.4: I can judge how strong the evidence is that the crust has moved and is still moving.	(personal)	(personal)	(personal)
ESS1-5.5: I can judge how well plate tectonics explains the ages of rocks, using evidence.	(personal)	(personal)	(personal)

Work plan

Retake CFA 5.5. Do the review for ESS1-5.3. Then the enrichment board.

Name _____ Period _____ Date _____

Review Day (continued)

Vocabulary check

Word	I can explain it to a partner (✓)	My example
lithosphere (plate)	✓	
plate boundary (divergent, convergent, transform)	✓	San Andreas Fault
mid-ocean ridge	✓	Mid-Atlantic Ridge
subduction	✓	Nazca plate
trench	✓	Mariana Trench
magnetic reversal	✓	Mirror stripes
craton (a continent's ancient core)	✓	Canadian Shield
impact crater	✓	Moon highlands

Before the CSA, I will study

My lowest targets. I will reread my notebook, redo the CFA questions I missed, and review the vocabulary.